

Research Assignment 1

BrightLearn Data Analytics

2026/09/09

Section A.: Data Fundamentals

- ① What is a Database? Explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

Database - Is an organized collection of structured information, usually stored electronically in a computer system.

Importance

- ↳ They keep systems secure
- ↳ The information kept/stored is accurate
- ↳ They store huge amounts of information
- ↳ Also serves as digital memory in the digital modern world

Examples (2)

↳ Uber

- By matching nearby drivers in real time, payment processing

↳ Instagram

- Maps out complex social webs, likes, posts, tags, mutual connections

- ② What is a Datawarehouse? How is it different from a regular database? And why would a company use a data warehouse?

Datawarehouse (DW) - Is a centralized storage system designed to specifically for data analysis (clean/collect), reporting (current and historical data) and business intelligence

Differences

- They differ in the way they store information. A regular data warehouse handles fast real-time daily transactions. Whereas, a data warehouse stores historical information for complex analysis.

Uses

- It is used for historical tracking and trend spotting
- Data quality and security
- It is best for decision making as all things in the business can be found in one place

③ What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse.

Data Lake

- It is a centralised 'vault' that stores massive or enormous amounts of data in its raw, original format

Differences

DW	DL
→ clean, organized data	→ raw original format (storage)
→ structured tabular data only	→ accepts info as it is
	→ structured, unstructured data

- A business would use a data lake over a data warehouse when its data needs expand beyond structured, predictable data into semi enormous volumes of raw unstructured or fast-streaming information.
- A Data Lake is more cost effective.

④ What is a data mart? How does it relate to a data warehouse and who typically uses it?

Data mart

- ↳ A data mart is a specialized section of a data warehouse focused solely on a single department e.g. marketing, sales and finance.

Relationships

- ↳ The data warehouse stores the entire company's history across all departments. A data mart stores only the metrics relevant to one specific subject.
- ↳ The central data warehouse collects, cleans and integrates raw data from across the whole business. The data mart then draws smaller, customized subset of that clean data to serve a specific team/department.

Who uses it

- ↳ Accountants
- ↳ Business Analysts
- ↳ HR & talent leads

⑤ Structured Data

- ↳ Data is more organized and well defined.

e.g. Inventory

Semi structured data

- ↳ Data is organized to some extent, a bit flexible.

e.g. server logs

Unstructured Data

- ↳ The data is not predefined to any structure or format

e.g. Images

- ⑥ What is data modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Data modelling

- ↳ Is a process of creating an illustration that maps out how data will be collected, organized, stored and connected within a system.

Importance

- ↳ Prevents duplication and ensures data quality - If there is no model, the different parts of an application often store duplicate information. This then leads to

data inconsistencies. Data modeling enforces rules and limitations to keep data or information clean and accurate.

- y A data model serves as a universal visual language. It ensures that software developers (coders or those who write code) and stakeholders agree on what the system will measure, store, process before development starts.

Two Data Models

Type 1 - The relational data model (SQL)

- y organises data into two dimensional tables consisting of columns and rows
- y The table link to each other using unique identifiers called primary and foreign keys.

Type 2 - The document data model

- y stores data in flexible, self-contained documents usually in JSON or BSON formats instead of fixed rows and columns.

⑦ What is Data mining? How ~~the~~ is it used in business? Provide two real world examples of data mining in action.

Data Mining

- y It's an automated process of digging through massive sets of data to discover hidden patterns, anomalies, correlating trends that humans cannot easily spot on their own.

Uses of Data Mining

- y Fraud & anomaly detection by spotting unusual activities
- y Customer segmentation - by grouping individuals based on shared ~~character~~ character traits without predefined labels.
- y Predictions - by analysing historical records to predict future outcomes

- ⑤ What is a Database Management System (DBMS). Name and briefly describe four popular DBMS platforms (e.g. MySQL, PostgreSQL, MongoDB, Oracle) and explain a suitable case for each.

A Database Management System

- ↳ Is a software that acts as a link between end users, software applications, and the physical database.
- ↳ It allows people to store and use programs, to organize, retrieve, update and secure data safely without needing to manually manage server storage files.

4 Popular DBMS Platforms

① PostgreSQL (Relational/SQL)

- ↳ An advanced open-source relational DBMS celebrated for strict adherence to SQL standards, high flexibility and total data integrity.

↳ Example: In banking platform, data precision is mandatory. PostgreSQL uses strict ACID compliance to guarantee ledger updates never fail mid-way or create duplicate entries.

② MongoDB (Document / No SQL)

- ↳ A leading NoSQL database that stores data as flexible, JSON-like documents rather than fixed tables with pre-declared rows and columns.

↳ Example: E-commerce product catalogs - online store items have different attributes (e.g. sizes, size, colour, style). MongoDB lets developers store these different attributes together in one collection without changing the database structure.

③ MySQL / Relational SQL.

- ↳ It is one of the most widely used open-source relational databases globally, valued for its speed, ease of management and widespread web hosting integration.
- ↳ Example: It powers millions of websites for example WordPress, because it integrates out of the box with standard web development stacks and handles heavy web receding traffic with ease.

④ Oracle:

- ↳ Is a multiple commercial relational database management system built for enterprise-grade workloads. It is known for, robust security, scalability, strict transactional compliance, it supports both SQL operations and JSON (modern workloads), spatial data and graph processing within a single engine.
- ↳ Example: Global retail banks processing credit and authorizations, account settlements and general ledger updates worldwide - systems stay online 24/7.

⑨ What is ETL? (Extract, Transform, Load). Explain each step and describe why this process is important in data analysis

ETL stands for Extract, Transform, Load. It is an automated data integration pipeline that collects raw data from multiple scattered systems, cleans and standardized / standardizes into a consistent format, and saves it into a central repository such as a Data Warehouse (DW), so it can be queried by analysts



Importance

- ↳ ETL enables long term trend analysis, forecasting and seasonal tracking by preserving historical context of the data.
- ↳ It also ensures high quality data by filtering out duplicates, corrupted or missing data in the transformation phase so that dashboards and business reports are built on accurate information.
- ↳ Consolidates all the data collected from all departments into a central place so finance, marketing, sales teams all work from the exact same numbers.

⑩ What are common data storage formats used in analytics (e.g. CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over the other/another.

↳ Firstly data storage formats dictate how data is encoded and structured on a disk. It is imperative to choose the right format, as it impacts analytical query performance, memory consumption, storage expenses and pipeline delays (latency).

① JSON (Javascript Object Notation) - Is a flexible, text based semi-structured format that stores data as key-value pairs & arrays.
↳ Usage - to export small spreadsheets, sharing data with non-technical users, ad hoc testing.

② CSV (Comma Separated Values) - A human readable, plain-text format where data is stored line-by-line as rows, with fields delimited by commas.
↳ Usage - Web KPI payloads, mobile application event logs, dynamic NoSQL data ingestion.

③ Apache Parquet - An open source, column oriented binary storage format designed specifically for fast analytical query engines.

→ Usage - Cloud data lakes, Snowflake, large-scale SQL analytics.

④ Apache ~~Avro~~ Avro - Is a compact row oriented binary format that stores data alongside a JSON-defined schema embedded in the file header

→ Usage - Realtime event streaming pipelines

⑪ What is artificial Intelligence? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

Artificial Intelligence - refers to the field of computer science dedicated to developing systems capable of performing tasks that typically require human cognitive abilities such as reasoning, solving problems and understanding language.

Difference between Narrow AI and General AI

<u>Narrow AI</u>	<u>General AI</u>
* Designed for one specific task * Operates within fixed parameters * Cannot adapt to unprogrammed fields * Exists everywhere today	* Flexible, human level intelligence * Learns & transfers skills to any area * Possess general reasoning capabilities * Theoretical (does not yet exist)
<u>* Example: Apple Siri, Google Lens</u>	* Example: JARVIS from Iron Man

⑫ What is Machine Learning (ML)? How does it differ from traditional programming
Describe the three main types of Machine Learning: Supervised, Unsupervised and Reinforcement Learning.

4. Machine Learning (ML) - is a branch of artificial intelligence (AI) where computers learn to make decisions or predictions by finding patterns in data, rather than following explicitly written human instructions.

Differences between ML and traditional programming

Machine Learning

- × Rule based with explicit instructions for solving problems.
- × Effective for tasks with well defined rules and inputs
- × Changes require manual updates to code

Traditional Programming

- × Data driven, with models learning patterns from data to make decisions
- × Handles complex tasks where rules are not explicitly defined.
- × Models adjust and improve automatically over time with new data



Three Main Types of Machine Learning

① Supervised Learning

- × Well defined goals
- × Reverse engineering
- × E.g. fraud or non-fraud transactions

② Unsupervised Learning

- × Outcome is based only on inputs
- × Outcome - usually clustering or segmentation

③ Reinforcement Learning

- × Start state and end state are defined
- × The agent discovers the path and the relationship on its own.

(13) What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how they work?

~~Data~~ Learn Deep Learning -

→ Is an advanced type of learning that uses multi-layered artificial brain networks to automatically find patterns in enormous amounts of data.

How they Relate.

→ Artificial Intelligence - the broad overall goal of making computers act, think solve problems like humans.

→ Machine Learning - A specific subset of AI where computers learn from data and examples instead of being hard coded with strict rules.

→ Deep Learning - Is a specialized subset of machine learning that uses "deep" networks with many layers to handle complex data like images, text, speech on its own.

What are Neural Networks?

Neural Networks - Artificial neural networks are computer systems inspired by the human brain. They are made of connecting or connected processing units called nodes or artificial neurons.

→ A standard neuron network has three main parts: input layer, hidden layer and output layer.

How do they work?

→ They work by passing data through layers and adjusting how they "think" when they make mistakes.

signals & weights [Data enters a node where it is multiplied by a weight] → Activation → Layer by Layer learning → Training and correction

- (14) What is Natural Language Processing (NLP)? Describe what it is and provide three real world (example) applications of NLP that you use in everyday life.

Natural Language Processing (NLP)

↳ It is a branch of artificial intelligence (AI) that helps computers understand, interpret, process and generate human language in both written and text and spoken voice - in a way that is naturally and contextually meaningful.

(3) Real world Applications of NLP

① Machine Translation

↳ Services like Google Translate move beyond simple word for word dictionary lookups.

↳ They use neural machine translation to analyze entire sentences structure, idiom phrases to generate fluent translations between languages.

↳ Everyday example: Street signs

② Smart voice assistants and speech recognition

↳ Tools like Google assistant, Apple Siri convert spoken sound waves into digital text using Automatic Speech Recognition (ASR)

↳ Everyday Example: Asking Apple Siri on your smartphone to play a song, call someone - it will check the specific database and respond naturally.

③ Auto-complete, spell checkers and smart reply

↳ Keyboard engines (like SwiftKey), email clients (Gmail smart compose) and writing tools (Grammarly) use predictive NLP models to anticipate the next word you want to type or suggest corrections for grammatical errors in real time.

↳ Everyday Example: Like typing "please find attached" in an email and the system auto suggests "file requested", or fixing a mistype word like "I documenta to definitely".

⑤ What is Large Language Model (LLM)? How is it different from standard Machine Learning (ML) model. Name two well known LLMs and explain what they are used for.

Large Language Model (LLM)

↳ Is a type of artificial intelligence designed to understand, process and generate human-like text. LLMs are trained on huge volumes of written data - such as books, articles, relationships and the context of language.

Standard Machine Learning Model

- × specialises in a single task, for example detecting spam, sales predictions etc.
- × Requires structured labeled data e.g., spreadsheets.
- × Numerical values, categories or specific scores
- × Must be retrained from scratch to handle a different problem.

Large Language Model

- × Capable of handling hundreds of tasks without retraining
- × Trained primarily on enormous unstructured text data
- × Natural human language (text inputs and text outputs).
- × Adapts to new instructions instantly

- Two well-known LLMs and their uses

① GPT-4 (by Open AI)

↳ one of the most capable multimodal LLM available, powering ChatGPT Plus and enterprise AI applications.

Uses

↳ Used for writing and editing complex documents, debugging code and translating language.

② Llama 3 (by Meta)

↳ An open weight LLM created by Meta designed freely be accessible to researchers and developers.

Uses

- ↳ Used by developers to build customized AI applications, power chatbots within social media platforms and conduct open AI research.

⑩ What is Generative AI? How does it work at high level? Name three tools that use Generative AI and describe what each one does.

~~Generative~~ Generative AI

- ↳ It is a branch of AI that creates new content - such as text, music, codes, images or audio, rather than just analysing or classifying existing data.

How does it work?

- ↳ Massive data ingestion & pattern learning - During its training phase the AI Model processes huge volumes of existing content. Rather than memorising the data, it learns the underlying patterns, structures and relationships between elements.
- ↳ Contextual processing - Modern generative tools rely heavily on neural network architectures like Transformers. These networks evaluate the context of your input prompt, paying attention to how words or visual components relate to one another.
- ↳ Probabilistic prediction & generation - When you give the AI a prompt, it calculates the most statistically likely output to follow that input. It generates new content piece by piece for text.

Three popular tools that use Generative AI

- ① GitHub Copilot - A generative coding assistant integrated directly into software development environments. It autocompletes lines of code.

writes entire functions, and suggest context-aware programming solutions in real-time as developers write code.

② Chat GPT - An interactive conversational assistant developed by OpenAI
↳ It generates natural human-like text to draft essays, summarize complex documents, write computer code, and brainstorm solutions based on text.

③ Midjourney - An AI powered image generation platform accessible via Discord. It converts natural language descriptive text prompts into detailed visual art, imagery and graphical designs.

④ What is Training Dataset? Why is the quality training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

Training Dataset

↳ It is a collection of information used to teach an AI or machine learning model how to recognize patterns and make decisions.

Why is the quality of Training Dataset important?

↳ Teaches the right rule - clean, accurate data teaches the model true patterns instead of random noise or mistakes.

↳ Sets the limit - Data quality sets the absolute ceiling on how smart of accurate AI can be; no amount of computer power can fix bad or messy input data.

↳ High quality data solves data ensures the AI performs well in the real world, rather than just memorizing the practice examples.

What could go wrong if data is incorrect or biased?

↳ Blindspot and breakdowns could happen if the dataset lack unusual scenarios, the AI would fail at the moment it encounters a normal real-world situation it was not prepped for.

↳ Unfair bias - If the training data leaves out certain groups or includes human prejudice, the AI will produce unfair results

↳ Wrong predictions - False or mislabeled information teaches the AI bad patterns, leading to incorrect answers and poor performance

(18) What is Computer Vision? How machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Computer Vision

↳ Is a field of AI that gives machines the ability to see, process and understand digital images and videos.

How machines 'see' and interpret images?

↳ A computer sees a picture not in the way humans do. It sees a digital image as a large grid of numbers representing the brightness and the colour of each tiny dot or pixel.

↳ The system identifies basic shapes like lines, circles and angles by correctly tracing edges.

↳ Learning features via AI, when you use machine learning and neural networks, the system reviews thousands of training images to recognize complex features such as wheels on a car or eyes on a face.

Two real-world uses

① Manufacturing and quality control - Systems use cameras on assembly lines to inspect manufactured goods in real time, instantly catching defects, broken parts or faulty packaging.

② Health care and medical imaging - Microsoft Azure notes that computer vision analyzes X-rays, CT scans and MRIs to help doctors spot diseases and make fast accurate diagnoses.

- ⑨ What is an API (Application Programming Interface). Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems.

Application Programming Interface (API)

↳ Is a set of rules and protocols that allows different software applications to communicate and share data with one another. It acts as an intermediate messenger, taking a request from one system, telling another system what to do and returning the response back.

A simple real-world analogy - The restaurant waiter

↳ Imagine you are sitting at a table with a menu in front of you at a restaurant.

* The client - You order food from the kitchen, because you cannot walk into the kitchen and make it yourself

* The kitchen - Has all the ingredients (raw, cooked), recipes and tools needed to prepare the meal

* The waiter (The API) - the messenger who bridges the gap. The waiter takes the order (API request), delivers it to the kitchen, waits for the meal to be prepared and brings the food back to your table (the API response).

Why APIs are important?

- ↳ Scalability - Large applications break down complex systems into smaller independent services connected via APIs
- ↳ Enhanced Security - an API creates a protective barrier between systems
- ↳ APIs enable automated data pipelines, allowing applications to sync updates in real-time

② What is an API key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

An API key

↳ It is a unique code used by computer programs to identify and authenticate a user, developer or specific software application. Example: a digital password presented to gain access.

What is the purpose of API keys

- ↳ Helps providers monitor usage, track data analytics, and manage billing or rate limits (e.g., limiting a free account to 1000 requests per day)
- ↳ Controls what data or features your application is allowed to access
- ↳ Proves to the receiving server that the application is who it claims to be.

How to store and protect

- ↳ Never leave the API key out in the open
- ↳ Store keys in environment variables (like a .env file) on your local server, rather than writing them directly into your source code.
- ↳ Use dedicated tools like AWS Secrets Manager for production apps or HashiCorp Vault.
- ↳ Restrict key permissions by configuring your API key to have the minimum access required.

Consequences of Public Exposure

- ↳ Data breaches
- ↳ Financial loss
- ↳ Service disruptions

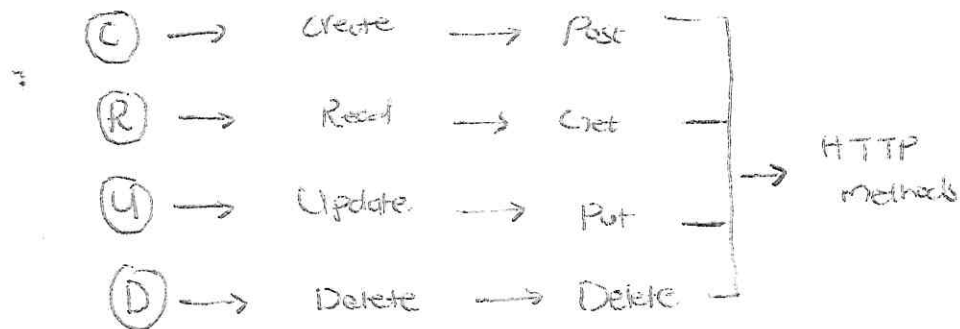
③ What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs. Briefly explain each method.

Rest API

It is an architectural style for designing networked applications that use standard web communications to interact with data.

REST stands for Representational State Transfer

HTTP methods



Main HTTP Methods in REST APIs

Rest APIs map standard database operations (CRUD: Create, Read, Update, Delete) to specific HTTP methods

Get (Read) - Retrieves data from server without modifying anything

- E.g., Fetching a list of products

Post (Create) - Sends data to the server to create a brand new resource

- E.g., Submitting a sign up form to create a new user account

Put (Update) - Sends updated data to the server to replace an existing resource

- E.g., Updating all contacts / details on an existing profile

Delete (Delete) - Removes a specific resource from the server

- E.g., deleting an unwanted file or user account

② What is JSON (Javascript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

JSON (Javascript Object Notation)

Is a lightweight, text-based data interchange format used to store and transmit structured data. It represents data in an easy-to-read format.

Why is JSON commonly used in APIs and data exchange?

- Unlike other formats like XML, JSON keeps file sizes small, leading to faster data transmission over networks and reduced bandwidth usage.
- It is straightforward, highly readable for developers, simplifying debugging and testing.
- Processes data efficiently.
- Flexible data structures.

Example of a JSON object

```
json
{
  "userId": 1024,
  "name": "Bardo Seleleko",
  "email": "seleleko b@gmail.com",
  "isAdmin": false,
  "skills": ["Python", "JavaScript"],
  "address": {
    "city": "Johannesburg",
    "state": "Gauteng"
  }
}
```

(23) What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a good poorly written prompt and an improved version of each.

Prompt Engineering

- ↳ It is the practice of designing, structuring given to Generative AI models so they produce the most accurate, relevant and helpful responses possible.

Why prompt engineering matters?

- ↳ Precise prompts give you useful answers on the first attempt rather than after multiple follow-ups.
- ↳ Techniques like assigning roles or step by step instructions allow models to handle complex analytical, coding or writing tasks.
- ↳ Ensures outputs match specific constraints like length, formatting, brand voice or audio recording level.

Examples: Poor vs. Improved Prompts

① Business Email.

- x Poor Prompt - "Write an email asking for a job."
- x Improved Prompt - "Act as a professional career counselor. Write a concise, polite follow up email (under 80 words) to a recruiter asking about the status of my application for a Data Analyst role. Use a warm, confident tone."

② Explaining a technical concept

- x Poor prompt: "Explain machine learning."
- x Improved prompt: "Explain machine learning in simple terms to a 10-year old child. Use a real-world analogy (like learning to play a sport). Limit the explanation to 3 short paragraphs and avoid technical jargon."

- ②④ What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or Javascript.

• A Map function

↳ applies a specific action or transformation to every item in a collection of data and return a new collection containing the modified results.

What it does in programming and data processing

- ↳ Transforms data: Changes every element in a list or array based on a rule you provide
- ↳ keeps original data safe: Leaves the original collection unchanged while building a new one.
- ↳ Processes efficiently: Handles large datasets quickly and cleanly in functional programming.

Python Example

```
# Original list of numbers
```

```
numbers = [1, 2, 3, 4]
```

```
# Apply map with a lambda function
```

```
squared = list(map(lambda x: x**2,
```

```
print(squared) # Output: [1, 4, 9,
```

- (2f) What is cloud computing? Name the three main cloud service model (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each support model.

Cloud computing :

↳ Is the delivery of computing services — including servers, storage, databases, networking, software and intelligence — over the internet ("the cloud") instead of local hard drives or physical data centers.

Three main cloud service models

① Infrastructure as a Service (IaaS)

→ Provides important computing infrastructure over the internet, such as virtual servers, storage and networking. You manage the operating system, data and applications while the provider manages the physical hardware.

E.g., Amazon EC2 (Elastic ~~Computer~~ Compute cloud)

② Platform as a Service (PaaS)

→ Supplies an on-demand, managed environment with tools to build, test, deploy and manage software applications.

E.g., Google App Engine

③ Software as a Service (SaaS)

→ Delivers fully functional, ready to use software applications over the internet, usually via a subscription. The provider handles all maintenance, updates etc.

E.g., Gmail.

②6) What is a Version Control System? What is a Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

Version Control System (VCS)

→ It is a software that tracks changes made to files over time. It allows you to recall specific historic versions later, compare updates and collaborate safely without overwriting other people's work.

Git

→ Is a free, open-source distributed version control system created by Linus Torvalds in 2005. Unlike other centralized systems, Git gives every user a complete local copy of the entire project history directly on their machine, allowing them to work offline and sync changes later.

Why Data analysts and developers use Git?

- Keeps a detailed record of who made changes, when it was made and why (via log messages)
- Multiple developers and/or data analysts can work on different parts of the same project simultaneously and merge their updates together safely.
- Eliminates endless file naming chaos
- Lets teams test new features, tools, algorithms in isolated environments without risking breaking the existing project.

What is a 'Commit' and a 'Branch'?

① Commit (A project snapshot)

- A saved snapshot of your files @ a specific point in time
- When you make progress on a script or analysis, you "commit" your changes alongside a short descriptive message. This permanently logs that milestone into the project history

② Branch (An independent workspace)

- A parallel line of development split off from the main project path.
- The primary codebase usually lives on a default branch (often named main). When starting a new feature or analysis task, you create a new branch

②⑦ What is Jupyter Notebook? What programming language is it mostly commonly used with? Why is it popular among Data Analysts and Data Scientists?

Jupyter Notebook

- Is a free-web based tool that lets you write code, add explanatory notes and view charts or data tables in a single interactive document.

Most Common Programming Language

- Python is the language most commonly used with Jupyter Notebooks
- While it supports other languages like R and Julia, Python is the top choice for data work.

Why It is popular among data analysts and data scientists

- You can save and send the entire notebook to teammates so they can see your data, your code and your final results all in one place.
- When you run a piece of code, the result or graph shows up immediately right below that specific cell.
- The document is split into boxes called "cells". You can run each cell of code individually instead of running the whole program at once.

28) What is Python? Why is it considered one of the most popular programming languages for Data Analysts and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python

- Is a high level, general purpose programming language designed to be easy to read and simple to write. Created with an emphasis on clean, English-like syntax.

Why Python dominates data analytics and AI

- Python easily connects with other systems and programming languages
- It allows developers to instantly load pre-written tools called libraries
- Its simple layout allows for beginners and non-programmers to learn it quickly and focus on solving data problems.
- Python easily connects with other systems

Four essential Python libraries for Data work

① Pandas - Organizes and manipulates structured data

Description - Think of it as Microsoft Excel inside Python. It lets you clean, merge, filter and analyze tabular (rows & column) data efficiently.

② NumPy - Handles complex mathematical calculations

Description - Short for 'Numerical Python' this library powers the heavy math behind data processing, making calculations fast.

③ Matplotlib - used for data visualization

Description - It turns raw numbers into visual insights. You can use it to build charts, graphs, histograms to easily map out trends and patterns.

④ Scikit-learn - Used for machine learning and predictive modeling

Description - The library provides simple tools to build artificial intelligence models.

②9 What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

Data Privacy

→ Is the right of individuals to control how their personal information is collected, used and shared by others.

→ It includes things like personal info, control (access) and ethics.

Why is it important?

→ Protects against unwanted tracking, spam etc.

→ Keep my details private out of the hands of criminals.

→ Helps people safe when using websites apps and stores.

What is the GDPR?

→ The General Data Protection Regulation (GDPR)

→ Is a strict privacy law created by the European Union (EU) that went into effect on May 25, 2018.

- x Forces organizations to be transparent about what data they take and keep it safe
- x Gives people the right to see their data, fix mistakes & ask companies to delete their info permanently.
- x Imposes penalties on companies that break the rules with

Which countries does it apply to?

→ Inside the EU - applies to all 27 member countries

→ Outside the EU - It also applies to any company or website in the world if they collect or process data from people who live in the EU.

③ What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Data ethics

- Is a branch of ethics that studies how data is collected, shared and used, ensuring that data practices do not harm people or violate their rights.

Three Examples of unethical uses of data or AI, along with why they are problematic:

① Biased AI Recruitment Tools

Example: A major tech company built an AI tool to automatically scan resumes and rank job applications

Why it's problematic: The AI automates and scales historical data human bias, unfairly penalizing qualified candidates based on gender rather than merit.

② Deepfakes used for misinformation

Example: Generative AI tools are used to clone a politician's voice and face, producing a highly convincing talk video showing them saying offensive things right before an election day.

Why its problematic: It manipulates truth, damages a person's reputation without their consent, weakens public trust in media or political institutions.

③ Covert Location Tracking

Example: A simple flashlight or weather app tracks a user's precise GPS location 24/7 in the background and sells the data to advertisers without the user's knowledge.

Why its problematic: It violates data privacy and informed consent for the user's end.

③ What is Data Governance? Define it and explain the key components of a good data governance framework.

Data Governance

It is a system of rules, roles and processes that ensure an organization's information is accurate, easy to use and safe.

The key components of a good data governance framework

① Tool and technology

* Concept - the software & systems used to help humans manage data

* The How - Companies use specialized data (software) called data catalogs to manually trace millions of pieces of data

② Security and Privacy

* Concept - protect sensitive information from being stolen, abused or leaked.

* The How - Not everyone who walks into the bank should have access to people's financial info. Permission, password should be enabled and info should be accessed on a need to know basis

③ Data Quality

Example / Concept - A process should be in place to ensure information is accurate, up to date and complete.

The How - Bad data leads to bad business decisions.

④ Roles & Responsibilities

Concept - Roles / Responsibilities should be assigned accordingly.

The How - Data owners and data stewards play in different roles as one is or would be the manager and the other day to day caretaker.

③② What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

→ Bias in AI means a computer system makes unfair or skewed choices that treat certain groups of people worse than others, as reported by IBM.

What it means for an AI to be biased

- If the past data contains human unfairness, the AI treats these bad habits as rules.
- Instead of being fair, the computer repeats and even grows the old unfairness.

A Real-World example

- A company builds an AI to read job applications and pick the best ones.
- The AI learns that being a man is a good trait for the job. It starts throwing away great resumes just because the applicant is a woman.

③③ What are the key differences between a Data Analyst, Data Scientist, a Data Engineer, and an AI engineer? Create a simple table comparing their roles, skills and tools.

- y A. Data Engineer, Data Scientist, Data Analyst and AI Engineer work with data, but their jobs are completely different.

Data Analyst - Looks @ past data to explain what happened

Data Engineer - Builds the plumbing and roads that move data from one place to another.

Data Scientist - Uses math and patterns to predict what will happen in the future.

AI Engineer - Build actual smart products using ready made AI tools.

Comparison Table

Role	What they do	Main Skill	Common Tools
Data Analyst	Helps business leaders make decisions	Basic math, data cleaning	Excel, SQL
Data Engineer	Builds and fixes pipe lines that keep data safe	Computer coding	SQL, Apache Spark
Data Scientist	Builds models that assisting in future forecasts	Advanced statistics	Python, R
AI Engineer	Connects software to AI models	Software building	Python

③+ What is business intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help business make data driven decisions.

Business Intelligence is like a dashboard in a car for a company. Just like a dashboard in a car it turns complex mechanical data into simple visuals - such as fuel gauge or speedometer.

Two Popular BI tools

→ Power BI - this tool is highly popular because it connects seamlessly with other popular Microsoft products like Excel.

→ Tableau - It is famous for its highly customizable visual charts.

- How BI tools help businesses make smart decisions
 - ↳ Finding and fixing mistakes
 - ↳ Tracking goals
 - ↳ Spots trends instantly.

35) Which concept did I find most interesting or surprising?

I found the concept of predictive analytics the most interesting. It surprised me because I always thought data was a boring topic. Learning that a business can look at old data and accurately predict exact customer trends is amazing. It completely changed how I view numbers, reports which are clues to the future.

36) Which concept did I find most difficult to understand?

The hardest concept for me was Data pipelines and understanding how data is cleaned and moved. To understand better I looked up videos on YouTube that used analogs (visual), comparing data pipelines, filtering and all. Breaking the technical terms down into everyday pictures made it easy.

37) How do I see Data & AI affecting my future career everyday life.

In my everyday life I already see AI and data quietly managing my routines. Like finding my way home everyday through voice apps, recommended music, apps. In my future career I expect to use smart dashboards to track my daily goals like BI tools. To avoid endless paperwork in my daily life, I can rely on data systems to instantly highlight areas that need attention/improvement which will enable me to make faster decisions without needing to be a computer expert.

1. Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling* (3rd ed.). John Wiley & Sons.
2. Inmon, W. H. (2005). *Building the Data Warehouse* (4th ed.). John Wiley & Sons.
3. **Watson, H. J., & Ariyachandra, T. N. (2006).** *Data warehousing governance: Best practices for success.* **Information Systems Management**, 23(1), 85–95.
4. Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson. Notes: The standard international textbook defining AI systems, agent architectures, and the distinction between domain-specific intelligence and general reasoning.
5. Goertzel, B., & Pennachin, C. (Eds.). (2007). *Artificial General Intelligence*. Springer.
6. Notes: A foundational collection defining "Artificial General Intelligence" (AGI) to distinguish flexible, human-level intelligence from specialized "Narrow AI."
7. McCarthy, J., Minsky, M. L., Rochester, N., & Shannon, C. E. (1955). A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence.
8. **Chen, P. P.-S. (1976).** *The entity-relationship model—toward a unified view of data.* **ACM Transactions on Database Systems (TODS)**, 1(1), 9–36. Notes: The landmark paper introducing the Entity-Relationship (ER) model, establishing how system data structures are visually mapped
9. **Simsion, G. C., & Witt, G. C. (2004).** *Data Modeling Essentials* (3rd ed.). Morgan Kaufmann.
10. Inmon, W. H. (2005). *Building the Data Warehouse* (4th ed.). John Wiley & Sons.
11. Notes: Explains the operational necessity of extracting data from transactional systems, transforming it into integrated structures, and loading it into analytical databases
12. Kimball, R., & Caserta, J. (2004). *The Data Warehouse ETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering Data*.
13. Vassiliadis, P. (2009). A survey of Extract–Transform–Load technology. *International Journal of Data Warehousing and Mining (IJDWM)*, 5(3), 1–27.
14. Notes: A comprehensive survey analyzing the core mechanics, optimization techniques, and structural phases of ETL workflows.
15. Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
16. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., ... Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877–1901. <https://arxiv.org/abs/2005.14165>
17. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008. <https://arxiv.org/abs/1706.03762>

18. Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27, 2672–2680. <https://arxiv.org/abs/1406.2661>
19. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008. <https://arxiv.org/abs/1706.03762>
20. White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., Elnashar, A., Gomez, J., Dai, A., Turner, M., & Schmidt, D. C. (2023). A prompt pattern catalog to enhance prompt engineering with ChatGPT. arXiv preprint. <https://arxiv.org/abs/2302.11382>
21. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824–24837. <https://arxiv.org/abs/2201.11903>
22. Chacon, S., & Straub, B. (2014). Pro Git (2nd ed.). Apress. <https://git-scm.com/book/en/v2>
23. Loeliger, J., & McCullough, M. (2012). Version Control with Git: Powerful tools and techniques for collaborative software development (2nd ed.). O'Reilly Media.